

E-Commerce Sales Analysis Report

1. Introduction

This report presents an analysis of e-commerce sales data using Python. The objective of the analysis was to understand sales and profit trends over time, identify the best- and worst-performing product categories, and evaluate the performance of different customer segments. The insights generated can help guide decisions around inventory planning, discounting strategy, and marketing focus.

2. Dataset Overview

The analysis was performed on the "Sample - Superstore" dataset, uploaded as a CSV file. The dataset contains **9,994 records** across **21 columns**, including order details (Order ID, Order Date, Ship Date, Ship Mode), customer information (Customer ID, Customer Name, Segment), location details (Country, City, State, Region), and product information (Category, Sub-Category, Product Name), along with the key numerical fields: **Sales**, **Quantity**, **Discount**, and **Profit**.

3. Methodology / Steps Followed

The analysis was carried out in Python using the **pandas** library for data handling and **Plotly** for visualization. The steps followed were:

1. **Imported the required libraries** — pandas for data manipulation and Plotly Express/Graph Objects for interactive visualizations.
2. **Loaded the dataset** from the CSV file (`Sample - Superstore.csv`) into a pandas DataFrame.
3. **Explored the data** using `.head()`, `.describe()`, and `.info()` to understand its structure, data types, and check for missing values (the dataset had no null values across any column).
4. **Converted the "Order Date" and "Ship Date" columns to datetime format** to enable time-based analysis.
5. **Created new derived columns** — "Order Month," "Order Year," and "Order Day of Week" — extracted from the Order Date, to allow grouping and trend analysis by time period.
6. **Performed grouped aggregations and visualizations** to analyze sales and profit across time, product categories, sub-categories, and customer segments (detailed below).

4. Analysis and Visualizations

4.1 Monthly Sales Analysis

A line chart was plotted to show total sales for each month (aggregated across all years in the dataset).

- Sales started relatively low in **January (~\$94,925)** and **February (~\$59,751)**, the weakest month overall.
- Sales picked up sharply from **March (~\$205,005)** onward, followed by a mid-year plateau between April and August (roughly \$137,000–\$159,000 per month).
- The strongest performance was seen in the last quarter of the year: **September (~\$307,650)**, **October (~\$200,323)**, **November (~\$352,461)** — the highest-selling month — and **December (~\$325,294)**.
- This pattern indicates a strong **seasonal, holiday-driven sales trend**, with the business generating a large share of its annual revenue between September and December.

4.2 Sales Analysis by Category

A pie chart was used to break down total sales across the three main product categories.

- **Technology:** ~\$836,154 (highest sales contributor)
- **Furniture:** ~\$742,000
- **Office Supplies:** ~\$719,047

Sales are fairly evenly distributed across the three categories, with Technology holding a slight edge.

4.3 Sales Analysis by Sub-Category

A bar chart broke sales down further into 17 sub-categories.

- Top performers: **Phones (~\$330,007)**, **Chairs (~\$328,449)**, **Storage (~\$223,844)**, **Tables (~\$206,966)**, and **Binders (~\$203,413)**.
- Lowest performers: **Fasteners (~\$3,024)**, **Labels (~\$12,486)**, and **Envelopes (~\$16,476)**.

This shows sales are concentrated in a handful of high-value sub-categories, while many smaller office-supply items contribute only marginally to total revenue.

4.4 Monthly Profit Analysis

A line chart was used to track profit trends month by month, mirroring the sales analysis.

- Profit followed a broadly similar seasonal pattern to sales, starting low in **January (~\$9,134)**, rising through the year, and peaking toward year-end: **September (~\$36,857)**, **November (~\$35,468)**, and **December (~\$43,369)** — the most profitable month.
- **March (~\$28,595)** also stood out as a strong profit month relative to its neighboring months.

4.5 Profit Analysis by Category

A pie chart broke down profit contribution by category — and revealed an important gap compared to the sales breakdown.

- **Technology:** ~\$145,455 profit (highest)
- **Office Supplies:** ~\$122,491 profit
- **Furniture:** only ~\$18,451 profit

Although Furniture generated nearly as much revenue as the other two categories (~\$742,000 in sales), it contributed disproportionately little profit — signaling weak margins in this category.

4.6 Profit Analysis by Sub-Category

A bar chart of profit by sub-category showed some sub-categories were actually **losing money**:

- **Loss-making sub-categories:** **Tables (~\$17,725)**, **Bookcases (~\$3,473)**, and **Supplies (~\$1,189)**.
- **Most profitable sub-categories:** **Copiers (~\$55,618)**, **Phones (~\$44,516)**, **Accessories (~\$41,937)**, **Paper (~\$34,054)**, and **Binders (~\$30,222)**.

This confirms that the weak Furniture profit seen above is being driven mainly by Tables and Bookcases, both of which sell reasonably well but erode profit — likely due to high discounting or high shipping/production costs.

4.7 Sales and Profit Analysis by Customer Segment

A grouped bar chart compared total sales and profit across the three customer segments.

Segment	Sales	Profit
Consumer	~\$1,161,401	~\$134,119
Corporate	~\$706,146	~\$91,979
Home Office	~\$429,653	~\$60,299

The Consumer segment is the largest contributor to both sales and profit, followed by Corporate and then Home Office.

4.8 Sales-to-Profit Ratio by Segment

To evaluate how efficiently each segment converts sales into profit, a Sales-to-Profit ratio was calculated (lower is better — it means less sales revenue is needed to generate \$1 of profit):

Segment	Sales-to-Profit Ratio
Home Office	7.13
Corporate	7.68
Consumer	8.66

Interestingly, although the **Consumer** segment generates the most total sales and profit, it is the **least efficient** at converting sales into profit. The **Home Office** segment, despite having the smallest sales volume, is the **most efficient**, generating relatively more profit per dollar of sales.

5. Key Conclusions

1. **Strong seasonality:** Sales and profit peak sharply in the last four months of the year (September–December), especially November and December, suggesting the business should plan inventory, staffing, and marketing campaigns around this seasonal surge.
2. **Technology is the best all-round performer:** It leads in both sales and profit, making it the most valuable category for the business.
3. **Furniture is a problem category:** Despite generating sales close to the other categories, its profit contribution is disproportionately low, mainly due to losses in the Tables and Bookcases sub-categories. These likely need a review of pricing, discounting, or cost structure.
4. **A few sub-categories are actively losing money:** Tables, Bookcases, and Supplies all show negative profit and should be prioritized for corrective action (e.g., reducing discounts or renegotiating supplier costs).
5. **Consumer segment drives volume, but Home Office is more efficient:** The Consumer segment brings in the most revenue and profit overall, but Home Office customers generate profit more efficiently relative to sales, which could make them a valuable segment to grow further.
6. **Revenue is concentrated in a few sub-categories:** Phones, Chairs, Storage, Tables, and Binders together account for a large share of total sales, while many smaller categories (Fasteners, Labels, Envelopes) contribute very little — useful to know for inventory and shelf-space prioritization.

6. Recommendations

- Reassess discounting strategy on **Tables and Bookcases** to restore profitability, since they sell well but currently generate losses.
- Double down on **Technology** and **Copiers/Phones**, which combine strong sales with strong margins.
- Plan for a **Q4 demand surge** each year by scaling inventory and marketing spend ahead of September–December.
- Investigate growth opportunities in the **Home Office segment**, given its superior profit efficiency compared to Consumer and Corporate.